

Reviewing AI-proposed pathway edges

A one-hour pass over 27 candidate relations. Expect to accept about five. The low yield is the point — it tells you the hand-built graph is already sound.

Where the file is. `atlas_data/relation_candidates_new.csv` in the Atlas project folder. Open it in Excel. It is UTF-8 with BOM, so accents render correctly.

What it contains. 194 candidate edges, auto-extracted from the word order in the `AI_Target` field across 275 studies. Sorted by `n_studies`, highest first.

What to actually review. Only the rows where `n_studies` ≥ 2 — the top **27 rows**. A single mention in free text is not a reason to draw an arrow on a mechanism diagram. Everything below that line can wait.

01 The one question you are answering

Did one of the cited studies actually demonstrate this step — or do the two molecules merely appear in the same sentence?

That is the whole job. The generator knows nothing about biology; it only noticed that two names sat next to each other, separated by a slash, in a field a language model wrote. Word order is a *hint* about direction of signal. It is not evidence.

So for every row you either open one of the cited papers and confirm the step, or you reject it and say why. There is no third option, and "it sounds right" is not a reason to accept — canonical textbook steps with no supporting study in the corpus belong in **Knowledge_Gaps**, not in Relations. That rule already sits in the Relations table description; this task does not change it.

02 Four ways the generator fails

These four patterns cover almost every bad row. Learn to spot them and the pass goes quickly.

A • Conjunction misread as a chain

The study covers *both* molecules; it does not claim one regulates the other. This is the most seductive failure because it produces the highest study counts.

`mTORC1 → mTORC2` (9 studies) · `TORC1 → TORC2` (3 studies) – reject both.

B • Sibling substrates

Both molecules sit *downstream of the same node*, not downstream of each other. They are listed together because one paper measured both readouts.

`p70 S6K → 4E-BP1` (3 studies) – both are mTORC1 substrates. Reject.

C · An alias of an edge you already have

Same claim, different name or direction. The graph already carries it. Adding it again creates a duplicate the Mechanism Explorer will draw twice.

mTOR → raptor is the existing Raptor required-for mTORC1 · AMPK → mTOR is AMPK → mTORC1 · SLC38A9 → Rag-Ragulator is SLC38A9 → Rag .

D · Transitive shortcut

True as a statement, but only *via* intermediate steps that are already drawn. Drawing the shortcut asserts a direct link no study demonstrated — and it quietly flattens the mechanism.

GATOR2 → mTORC1 — real, but it runs GATOR2 → GATOR1 → Rag → mTORC1, and all three of those edges exist. Reject the shortcut.

03 What to fill in

COLUMN	WHAT GOES THERE
DECISION	accept or reject . If rejecting, add the letter of the failure pattern — reject A , reject C . Takes two seconds and is worth more than a sentence.
SIGN	Only if accepting. One of: activates · inhibits · binds · required-for · recruits · no-effect. Use <i>required-for</i> when the source is permissive but not itself the switch.
MECHANISM	Only if accepting. One plain sentence saying how the source acts on the target, written so a motivated high-school reader follows it. No jargon you would not explain aloud.
DIRECTNESS	Only if accepting. How directly the causal link was shown. A cryo-EM structure is <i>direct</i> and still tier D for any human-health claim — the two axes are independent.

Leave everything else alone. `n_studies` , `source` , `target` and `study_sids` are provenance and should stay exactly as generated.

04 Two worked rows

mTORC1 → TFEB · 4 studies · MAR2012, NAP2020, ROC2012, SET2012

ACCEPT The clearest genuine gap in the whole file. mTORC1 phosphorylates TFEB on the lysosomal surface and keeps it out of the nucleus; when mTORC1 goes quiet, TFEB moves in and switches on the autophagy-lysosome gene programme. Four independent studies, and the edge is simply missing from the graph.

SIGN
inhibits

MECHANISM
Active mTORC1 tags TFEB with a phosphate at the lysosome, which traps TFEB in the cytoplasm — so as long as mTORC1 is on, the cell's autophagy genes stay off.

DIRECTNESS
direct

REJECT A Highest count in the file and still wrong. These nine papers compare the two complexes or measure both; none claims mTORC1 regulates mTORC2. The count is high precisely *because* the pairing is a common way to write about mTOR — which is the trap.

05 Why the rejections matter more than the accepts

Every rejection with a pattern letter is a datapoint about *where machine-proposed biology goes wrong*. After twenty or thirty of them the tally stops being anecdote and becomes a measurement: you will be able to say "conjunction errors are 40% of false proposals" with a number behind it.

That number is the prerequisite for the next thing the Atlas wants to do — automated contradiction detection. You cannot trust a machine to tell you which studies disagree until you know how it fails when it tells you which studies connect. The Relations table has a `Review_Flag` field built for exactly this tally.

So: do not delete the rejected rows. They are the more valuable half of the output.

06 What to expect

CANDIDATES GENERATED	COUNT	WHAT TO DO
Total proposed from AI_Target word order	230	—
Already present in Relations	36	filtered out for you
Backed by a single study	167	skip for now
Backed by ≥ 2 studies	27	your worklist
Realistic accepts	~5	most fail on patterns A-D

Five new edges on top of 43 is not a disappointing return. It means the hand-built graph already covers the backbone of the pathway, and that the four studies behind `mTORC1 → TFEB` were sitting in the corpus unconnected. Finding that was worth the hour.

A note on one row you will meet: `LC3B → p62` claims two studies, VER2026 and VERMA2026. Those were the same paper entered twice; the duplicate was removed on 27 July, after this file was generated. Treat it as a single study and skip it.

Generated 27 July 2026 from `normalize_entities.py` over 275 curated studies · Candidate edges carry status = Proposed and confidence = heuristic-order; word order in free text is a hint about direction, never evidence for it.